



CSCI 723 - Advanced Parallel Systems and GPU Programming

Жетілдірілген параллельді жүйелер және GPU бағдарламалау

Продвинутые параллельные системы и программирование на GPU

General Course Information

School: School of Engineering

Course Type: Elective Course

Semester in which the course is offered: Fall & Spring

Course instructor(s): Ben Tyler

Talgat Manglayev

Course Description - English

This course is intended for PhD students interested in the efficient use of modern parallel systems. Topics such as parallel computer architecture, programming models, memory hierarchy, parallel program design and parallel programming tools for multi-core systems and general-purpose graphics processing units (GPGPUs) will be covered. This comprehensive overview will equip students with a broad understanding of the key approaches in heterogeneous programming. Students will engage in practical, hands-on projects utilizing real-world scientific models to tackle contemporary challenges and applications, including physics, chemistry, biology and trustworthy machine learning. In the second part, the course covers the most common and current GPU parallel programming techniques with lab-based programming assignments using CUDA API with C/C++ and Python.

Course Description - Kazakh

Бұл курс қазіргі заманғы параллель жүйелерге қызығушылық танытатын PhD студенттерге арналған. Параллельді компьютер архитектурасы, бағдарламалау модельдері, жад иерархиясы, параллельді бағдарламаларды жобалау және көп ядролы жүйелер мен жалпы мақсаттағы графикалық процессорлар (GPGPU) үшін параллель бағдарламалау құралдары сияқты тақырыптар қарастырылады. Мұндай жаппай шолу арқылы біртекті бағдарламалаудың негізгі түсініктері шәкірттер бойларына кемел сіңеді. Студенттер физика, химия, биология және жасанды

интеллект салаларындағы заманауи мәселелерді шешуге бағытталған шынайы ғылыми модельдерді қолданатын жобалармен практикалық жұмыстарын өткізеді. Курстың екінші бөлімі CUDA API арқылы зертханалық бағдарламалау тапсырмалары бар GPU параллельді бағдарламалаудың ең кең таралған және ағымдағы әдістерін қамтиды.

Course Description - Russian

Этот курс предназначен для студентов обучающихся по программе докторантуры PhD, интересующихся эффективным использованием современных параллельных систем. Будут рассмотрены такие темы, как архитектура параллельных компьютеров, модели программирования, иерархия памяти, проектирование параллельных программ и инструменты параллельного программирования для многоядерных систем и графических процессоров общего назначения (GPGPU). Этот всесторонний обзор обеспечит студентов широким пониманием ключевых подходов в гетерогенном программировании. Студенты будут участвовать в практических проектах, используя реальные научные модели для решения современных задач и приложений, включая физику, химию, биологию и надежное машинное обучение. Во второй части курса рассматриваются наиболее распространенные и актуальные техники параллельного программирования на GPU с лабораторными заданиями по программированию, использующими API CUDA с C/C++ и Python.

Course Requisites

Conditions of Enrollment

Course Aims

- 1. To introduce students to concepts, hardware architectures and software programming models of parallel systems
- 2. To facilitate students' knowledge and understanding of parallel programming technologies
- 3. To provide students hands-on experience in designing and implementing simple parallel programs

Course Learning Outcomes (CLOs)

Course Learning Outcomes	Graduate Attributes
1. Define concepts related to CPU and GPU hardware architectures from mobile multi-core to server/cluster many-core parallel systems.	GA1 - Possess an in-depth and sophisticated understanding of their domain of study. GA2 - Be intellectually agile, curious, creative and open-minded. GA3 - Be thoughtful decision makers who know how to involve others. GA8 - Be prepared to take a leading role in the development of their country.
2. Articulate the differences of diverse parallel systems with the connection of parallel programming tools/APIs	GA1 - Possess an in-depth and sophisticated understanding of their domain of study. GA2 - Be intellectually agile, curious, creative and open-minded.

3. Critically assess the performance of various CUDA applications, utilizing both quantitative metrics and qualitative insights, to identify the bottlenecks;	GA4 - Be entrepreneurial, self-propelling and able to create new opportunities. GA5 - Be fluent and nuanced communicators across languages and cultures. GA6 - Be cultured and tolerant citizens of the world. GA8 - Be prepared to take a leading role in the development of their country.
4. Design and implement parallel programs using the parallel programming techniques and CUDA	GA1 - Possess an in-depth and sophisticated understanding of their domain of study. GA2 - Be intellectually agile, curious, creative and open-minded. GA6 - Be cultured and tolerant citizens of the world.
5. Integrate current research findings, methodologies, and advancements into heterogeneous computing	GA3 - Be thoughtful decision makers who know how to involve others. GA5 - Be fluent and nuanced communicators across languages and cultures.

Mapping of Major Core and Elective Courses

Program: Doctor of Philosophy in Computer Science (Active)

Course Learning Outcomes	Program Learning Outcomes
1. Define concepts related to CPU and GPU hardware architectures from mobile multi-core to server/cluster many-core parallel systems.	2. Identify appropriate research topics in their field of study in Computer Science, and generate hypotheses about such topics.
2. Articulate the differences of diverse parallel systems with the connection of parallel programming tools/APIs	2. Identify appropriate research topics in their field of study in Computer Science, and generate hypotheses about such topics.
3. Critically assess the performance of various CUDA applications, utilizing both quantitative metrics and qualitative insights, to identify the bottlenecks;	2. Identify appropriate research topics in their field of study in Computer Science, and generate hypotheses about such topics. 6. Develop their academic/scholarly career through presentations, publications, and national and international networking.
4. Design and implement parallel programs using the parallel programming techniques and CUDA	2. Identify appropriate research topics in their field of study in Computer Science, and generate hypotheses about such topics. 6. Develop their academic/scholarly career through presentations, publications, and national and international networking.
5. Integrate current research findings, methodologies, and advancements into heterogeneous computing	2. Identify appropriate research topics in their field of study in Computer Science, and generate hypotheses about such topics. 6. Develop their academic/scholarly career through presentations, publications, and national and international networking.

ECTS and Student Workload

Credits: 6

Contact Hours: 45

Self-Study Hours: 105

Total Hours: 150

Assessment Methods

Assessment Methods	Course Learning Outcomes (Reference #)
Midterm Exams (Two)	1, 2, 3, 4, 5
Quizzes	1, 2, 3, 4, 5
Lab and Homework Assignments	1, 2, 3, 4, 5

Assessment Methods Weighting

Assessment Methods	Weighting
Midterm Exams (Two)	40%
Quizzes	30%
Lab and Homework Assignments	30%
Total weighting	100%

Course Outline

Session Number	Theme (Key Topic)	Course Learning Outcomes (Reference #)
1	Course Overview. Introduction to parallel systems	1, 2
2	Fundamentals of Accelerated Computing with CUDA C/C++	2, 3, 4, 5
3	Fundamentals of Accelerated Computing with Modern CUDA C++	2, 3, 4, 5
4	Fundamentals of Accelerated Computing with Modern CUDA C++	2, 3, 4, 5
5	Optimize CUDA programs using Nsight profiling tools.	1, 2, 3, 4, 5

6	CUDA C++ features. Reduction algorithm implementation variations using CUDA C/C++.	1, 2, 3, 4
7	Review and Midterm	1, 2, 3, 4, 5
8	Fundamentals of accelerated computing with CUDA python. Introduction to CUDA with Numba. Custom CUDA Kernels in Python with Numba.	1, 2, 3, 4
9	Effective Use of the Memory Subsystem. Nsight Analysis System: Build Custom Python Analysis Scripts.	1, 2, 3, 4
10	Fundamentals of Accelerated Data Science.	3, 4
11	Fundamentals of Accelerated Data Science.	3, 4
12	Fundamentals of Deep Learning with Python calling CUDA functions.	1, 2, 3, 4
13	Fundamentals of Deep Learning with Python calling CUDA functions.	1, 2, 3, 4
14	Research issues on recent parallel systems. Summary and Course Review	1, 2, 3, 4

Learning and Teaching Methods

Course Learning Outcomes	Learning and Teaching Methods
1. Define concepts related to CPU and GPU hardware architectures from mobile multi-core to server/cluster many-core parallel systems.	Laboratories Lectures Homeworks; Quizzes; Exams
2. Articulate the differences of diverse parallel systems with the connection of parallel programming tools/APIs	Lectures Laboratories Homeworks; Quizzes; Exams
3. Critically assess the performance of various CUDA applications, utilizing both quantitative metrics and qualitative insights, to identify the bottlenecks;	Laboratories Lectures Homeworks; Quizzes; Exams
4. Design and implement parallel programs using the parallel programming techniques and CUDA	Lectures Laboratories Homeworks; Quizzes; Exams
5. Integrate current research findings, methodologies, and advancements into heterogeneous computing	Laboratories Lectures Homeworks; Quizzes; Exams

Laboratory Content

During labs, students are asked to solve tasks relevant to a new topic presented to them. The tasks have prepared code and students need to solve the remaining part of the task. Students are invited one by one to the teacher’s PC, which is projected to the class, to solve a task. The other students may either work on the task independently or follow the selected student’s solution. After completing the task, the student explains their approach to the class.

The students complete approximately one NVIDIA workshop every two weeks, where any remaining tasks are completed at home. In this way, students can earn a certificate of completion from NVIDIA, which they need to upload to receive a grade for their lab.

E-Learning

The course is not designed as an online course, but will make use of some online materials.

Online platform learning.nvidia.com Deep Learning Institute of NVIDIA. The online courses are step-by-step tutorials which have tasks at the end of each course.

Grading

Graduate

Letter Grade	Grade (quality points)	Grade Descriptors
A	4.00	Excellent; exceeds the highest standards in the assignment or course
A-	3.67	Excellent; meets the highest standards for the assignment or course
B+	3.33	Very good; meets high standards for the assignment or course
B	3.00	Good; meets most of the standards for the assignment or course
B-	2.67	More than adequate; shows some reasonable command of the material
C+	2.33	Acceptable; meets basic standards for the assignment or course
C	2.00	Acceptable; meets some of the basic standards for the assignment or course
C-	1.67	Acceptable; while falling short of meeting basic standards in several ways
F	0	Failing; very poor performance

Key Books and Learning Materials

Textbook				
#	Author(s):	Title	Publisher	Year

1	Wen-mei W. Hwu, David B. Kirk and Izzat El Hajj	Programming Massively Parallel Processors: A Hands-on Approach	Morgan Kaufmann	2026
2	Jason Sanders, Kandrot Edward and Jack Dongarra	CUDA by Example	Upper Saddle River, N.J: Addison-Wesley	20
3	John L. Hennessy and David A. Patterson	Computer Organization and Design 4th Edition (Ch. 7)	MK Publications	
4	John L. Hennessy and David A. Patterson	Computer Architecture: A Quantitative Approach 5th Edition (Ch. 3 and Ch. 4)	MK Publications	

Other Learning Materials

CUDA C Programming Guide (E-resources)

Academic Integrity Statement

[Student Code of Conduct and Disciplinary Procedures](#)

While general guidelines regarding academic integrity are published in the NU Student Handbook and by the school, for this course you must adhere to the following rules:

- You are not allowed to distribute or indirectly make your work available to others, even if you are not intending them to copy it -- this is still considered academic misconduct.
- While it is okay to talk through high-level concepts and ideas with your fellow students and to study with them, you are not allowed to give or receive direct help from others on graded assignments.
- Using code that you find online is strictly forbidden, as is using code generation tools such as ChatGPT, MS Copilot, Github Copilot, etc.
- Any appearance of cheating or inappropriate copying should be avoided, as such instances will be investigated and reported to the school.
- You may only get help on graded work from designated people—the instructors or TAs for the course. If you are struggling with something, by all means, please seek help from them.

In the event that academic misconduct such as plagiarism or cheating is discovered, the student will receive no credit for the work, and the event reported to your school. Egregious cases, or a second offense, can result in failure of the course and potential suspension or expulsion from the university.

When a student suspects that another student has violated the academic honesty policy, a report should be made to the appropriate faculty member.

Course Expectations

The Nazarbayev University Special Learning Needs Committee (SLNC) is committed to creating an equitable and inclusive education environment for all students. If you have a qualified special learning need (physical, cognitive, socio-emotional, and psychological), please contact the SLNC as early as possible to ensure you receive the fullest support available.

For more information, please contact us at SLNC@nu.edu.kz and follow this [link](#) to the Accommodation Request Instructions.

Course Expectations

Attendance

Missing classes and habitual tardiness will have a negative effect on your grade, both directly and indirectly. You are also responsible for any announcements made during the class period, so be sure to ask your instructor, TAs, or classmates for any info that you may have missed if you did not attend.

Electronic Resources

Labs will be conducted in one of our hybrid computer labs, which are designed to accommodate the full range of course activities. The necessary programming tools are installed on the classroom lab computers, and are available online for free download to your own computer. You are expected to check your Nazarbayev University e-mail on a daily basis for updates and announcements about the course.

Assignment Submission and Late Policy

Assignments must be submitted by the announced due date and time, as directed by the instructor. Some assignments may need to be submitted in the form of physical hard-copy in class, generally within the first five minutes of the start of the class period. Other assignments will need to be submitted digitally to Moodle. In case Moodle does not work, assignments need to be submitted by email to your instructor AND teaching assistant by the deadline date and time. In general, there is no late policy; if you submit an assignment after it is due, you get zero points for your assignment. In

cases of illness or family emergency, you must inform your instructor immediately if you believe you will not be able to submit your assignment on time. In such cases, an exception may be made at the discretion of your instructor.

Classroom Behavior

You are expected to act respectfully towards your fellow classmates, TAs, and instructors inside and outside of the classroom. We have a limited amount of time to cover a lot of material this semester, so you need to pay attention in class, and do your in-class work when it is assigned. Talking on your phone, texting, chatting online, browsing Facebook or other social media sites, and talking excessively with your neighbors about non-class related stuff are just a few examples of behavior that is not acceptable, and will negatively impact your grade.

Grade Policy

The passing grade for the course is D and above. As per CS department policy, students with failing marks within 5% of the passing threshold do not have the option of receiving an Additional Assessment Task at the end of the semester to increase their grade.

Supporting Documents

Course Approval Date

School Learning and Teaching Committee Approval:

Course Origination Date: 2026-06-16

Effective from: